Finding a Fire with Cameras

How to turn photographs into a location on a map

The problem. A wildfire starts somewhere in a valley. Several cameras on hilltops can see it. Each camera takes a picture. From those pictures alone, where exactly is the fire — and how big is it?

The catch. A photograph does not record distance. A small fire nearby and a huge fire far away can make the exact same picture. What a photo *does* record, very accurately, is a **direction**.

The solution, in one sentence. If two cameras in different places each measure the direction to the fire, we can draw two lines on a map and see where they cross — and that crossing point is the fire.

That is the whole idea. The rest of this guide is just doing it carefully, in three steps:

Part 1 Turn one *pixel* in one photo into a *direction* on a map. (Right-triangle trigonometry.)

Part 2 Cross the directions from *two* cameras to get a point. (Two equations, two unknowns.)

Part 3 Handle *three or more* cameras, which never quite agree, and find the best answer. (Careful averaging.)

What you need to know already. Not much:

- Right triangles: $\tan(\text{angle}) = \dfrac{\text{opposite}}{\text{adjacent}}$, and the inverse button $\tan^{-1}$ (also written arctan) on your calculator.
- Plotting a point (x, y), and the equation of a line $y = mx + b$ where m is the slope.
- Solving two equations in two unknowns.
- Squares and square roots.

No calculus. No matrices. Every formula is derived, and every number in the examples is worked out so you can check it on a calculator. Set your calculator to **degrees**.

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1 The big idea: a camera is a protractor

Imagine you and a friend are standing in a field, 100 metres apart, and a hawk is circling somewhere in front of you. Neither of you can tell how far away the hawk is. But each of you can point at it.

Now draw this on paper. Mark where you stand, and draw a line in the direction you are pointing. Do the same for your friend. The two lines cross at exactly one place — and that is where the hawk is. Neither of you knew the distance, yet together you found it.

A camera measures angles, not distances. Think of a camera as a very precise protractor that happens to record what it sees. It can tell you *which way* something is to better than a tenth of a degree. It can never tell you *how far*.

Two protractors in different places, both pointing at the same thing, pin down a location. This is called **triangulation**, and it is how surveyors, sailors, and fire lookouts have worked for centuries.

Everything that follows is bookkeeping on top of that picture: how to get the angle out of a photograph (Part 1), how to find where two lines cross (Part 2), and what to do when a third camera disagrees slightly (Part 3).

2 Setting up the map

Before any math, we need a map with coordinates on it.

2.1 Positions

We use ordinary (x, y) graph paper laid flat over the landscape, measured in metres:

$$x = \text{how far East}, \qquad y = \text{how far North}.$$

Everything in this guide is *flat* — a bird's-eye view, like looking straight down at a map. We ignore hills and heights. That keeps the math to two dimensions and is accurate enough for locating a fire.

2.2 Directions

A direction is an angle. We measure angles the way they are measured in math class: starting at the $+x$ axis (East) and turning **counter-clockwise**. So:

$$0^\circ = \text{East}, \quad 90^\circ = \text{North}, \quad 180^\circ = \text{West}, \quad 270^\circ = \text{South}.$$

We will call this angle θ ("theta"), and we call it the **bearing**.

Careful — compasses are different. A navigation compass measures angles *clockwise from North*. That is the opposite spin and a different starting line. If you need a compass azimuth β at the end, convert:

$$\beta = (90^\circ - \theta) \text{ adjusted to be between } 0^\circ \text{ and } 360^\circ.$$

Mixing up these two conventions is one of the most common ways to get a confident, completely wrong answer.

2.3 What we know about each camera

For each camera we need five numbers. Three describe *where it is and where it looks*; two describe *its lens*.

Symbol	Name	What it means, and how you get it
$(x_c,\ y_c)$	position	where the camera tower stands, from a GPS survey
ψ	heading	the bearing of the direction the camera points (the middle of its picture)
f	focal length	how "zoomed in" the lens is, measured **in pixels** (explained in §3.2)
c_u	centre column	the pixel column at the exact middle of the image, usually half the image width
W	image width	how many pixels wide the picture is

ψ is the Greek letter "psi". It is just a name for the camera's aim.

2.4 The example we will use throughout

Our example site. Three identical cameras. Each takes pictures $W = 1920$ pixels wide, and each lens sees 60° of the world from left edge to right edge. The centre column is $c_u = 1920/2 = 960$.

	Camera 1	Camera 2	Camera 3
Position (x_c, y_c), in metres	(0, 0)	(300, −400)	(900, 500)
Heading ψ	20°	70°	220°

A fire is spotted. In each photo, the smoke occupies a range of pixel *columns* — from its left edge to its right edge:

Camera 1: columns 1100 to 1400, Camera 2: 700 to 900, Camera 3: 880 to 950.

Those six numbers, plus the table above, are *all* we have. By the end we will know where the fire is to within about half a metre, and how many hectares it covers.

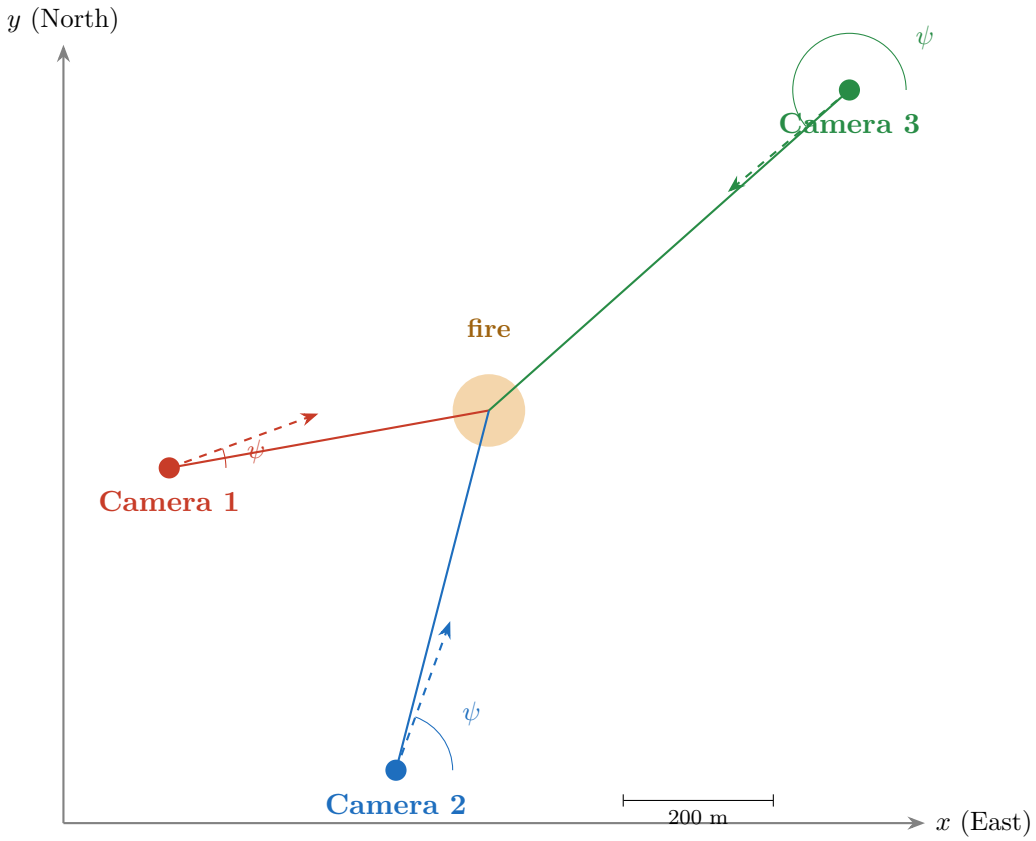


Figure 1: The example site, seen from above like a map. Dashed arrows show where each camera is aimed (its heading ψ). Solid lines are the directions to the fire that we are about to calculate. Notice that no camera knows the *distance* along its line — only the direction.

3 Part 1 — From a pixel in a photo to a direction on the map

3.1 How a camera really works (and why it's one triangle)

Inside a camera, light from the world passes through a tiny opening (the centre of the lens) and lands on a flat sensor. That is it. The whole of Part 1 comes from looking at this from above and noticing a right triangle.

Figure 2 shows the view from above. Put the opening at the point O. The sensor is a flat line at distance f in front of it. (The real sensor is *behind* the opening and the image is upside-down, but drawing it in front gives the same angles and avoids the flip.)

Look at the red triangle. Its two perpendicular sides are:

- the **base**, of length f — from the opening out to the sensor;

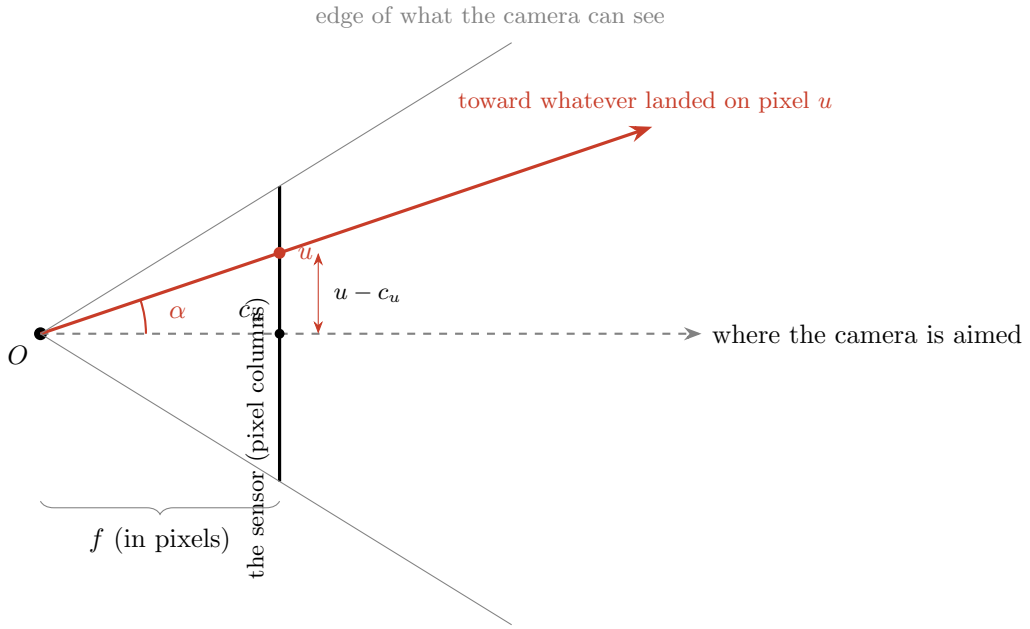


Figure 2: A camera from above. Light from something out in the world travels in a straight line through the opening O and lands on pixel column u. The shaded right triangle has height $u - c_u$ and base f, so its angle α satisfies $\tan\alpha = (u - c_u)/f$. **That triangle is all of Part 1.**

- the **height**, of length $u - c_u$ — how far the pixel sits from the middle of the image.

Both lengths are measured in *pixels*, which feels strange but is exactly right: what matters is their *ratio*, and the units cancel.

3.2 Step 1 — Find the focal length in pixels

The focal length f says how zoomed in the lens is. A long lens (big f) magnifies a narrow slice of the world; a wide lens (small f) squeezes a lot into the picture.

If nobody has measured f for your camera, you can compute it from the **field of view** — the total angle the camera sees, left edge to right edge. Call it FOV. Look again at Figure 2: the triangle from the centre to the *edge* of the sensor has height $W/2$ (half the image width) and base f, and its angle is half the field of view. So $\tan(\text{FOV}/2) = (W/2)/f$, and solving for f:

$$\boxed{f \;=\; \frac{W/2}{\tan(\text{FOV}/2)}} \tag{1}$$

Our cameras. $W = 1920$ pixels and $\text{FOV} = 60°$, so

$$f \;=\; \frac{1920/2}{\tan(60°/2)} \;=\; \frac{960}{\tan 30°} \;=\; \frac{960}{0.57735} \;=\; 1662.77 \text{ pixels.}$$

3.3 Step 2 — Turn the pixel into an angle

Now use the triangle. The angle α ("alpha") between the camera's aim and the direction to our object satisfies $\tan\alpha = (u - c_u)/f$. Undo the tangent with $\tan^{-1}$:

$$\boxed{\alpha \;=\; \tan^{-1}\!\left(\frac{u - c_u}{f}\right)} \tag{2}$$

If the object is right in the middle of the picture, $u = c_u$, the top of the fraction is zero, and $\alpha = 0$ — it is straight ahead, as it should be. If it is to the right, $u > c_u$ and α is positive.

The tempting shortcut that is wrong. It looks like you could just say "the picture is 1920 pixels wide and covers 60°, so each pixel is 60/1920 of a degree":

$$\alpha \overset{?}{\approx} (u - c_u) \times \frac{\text{FOV}}{W}.$$

This does not work, except very near the centre. The sensor is a *flat line*, so equal steps in pixels are equal steps in *distance*, not in *angle*. Out near the edge of the picture, a pixel covers less angle than one in the middle — the same reason a flat map distorts near its edges.

How wrong is it? For our camera:

pixel u	correct α	shortcut	error	error at 5 km
1100	4.8128°	4.3750°	−0.44°	38 m
1250	9.8933°	9.0625°	−0.83°	72 m
1400	14.8218°	13.7500°	−1.07°	94 m
1700	23.9910°	23.1250°	−0.87°	76 m

Being 93 metres off is the difference between finding a fire and driving past it. **Always use** $\tan^{-1}$**.**

3.4 Step 3 — Add the camera's heading to get the bearing

The angle α is measured *from the camera's own aim*. To put it on the map we add in where the camera is pointing:

$$\boxed{\theta \;=\; \psi \;-\; \alpha \;=\; \psi - \tan^{-1}\!\left(\frac{u - c_u}{f}\right)} \tag{3}$$

This θ is the **bearing**: the direction from the camera to the object, measured on the map, counter-clockwise from East.

Why minus and not plus? Because the two angles spin in opposite directions. In a photograph, pixel numbers grow toward the **right**. But map bearings grow **counter-clockwise**, which is toward the **left**. So something further right in the picture has a *smaller* bearing — hence the minus sign.

Check this once, for real. Point the camera at a landmark whose coordinates you know — a radio mast, a road junction. Compute its bearing with (3) and compare with the bearing you get from the map. If they disagree, you have a sign flipped somewhere. Do this every time anything in the camera setup changes.

3.5 Step 4 — The direction becomes a line on the map

We now know the fire lies somewhere along a particular straight line starting at the camera. Two equivalent ways to write it:

As a walk.

Start at the camera and walk a distance r in direction θ:

$$x \;=\; x_c + r\cos\theta, \qquad y \;=\; y_c + r\sin\theta, \qquad r > 0. \tag{4}$$

As a line through a point with a slope.

The slope of a line at angle θ is $m = \tan\theta$, so

$$\boxed{y - y_c \;=\; m\,(x - x_c), \qquad \text{where } m = \tan\theta.} \tag{5}$$

The second form is the familiar one from algebra class, and it is what we will solve with in Part 2.

The one thing a single camera can never tell you. In (4), the distance r is not determined — *any* positive value gives a point on the line. One camera gives you a direction and nothing more. To pin down r you need a second camera. That is Part 2, and it is the entire reason this problem is interesting.

3.6 A fire is not one pixel: it becomes a wedge

Real smoke covers a range of columns, from a left edge u^L to a right edge u^R. Run *each* edge through (3) and you get two bearings instead of one. Because bigger u means smaller θ:

$$\theta^{\max} = \theta\big(u^L\big) \quad \text{(left edge of photo)}, \qquad \theta^{\min} = \theta\big(u^R\big) \quad \text{(right edge of photo)}. \tag{6}$$

The fire lies somewhere in the pie-slice between those two bearings. We call that slice a **wedge** (Figure 3).

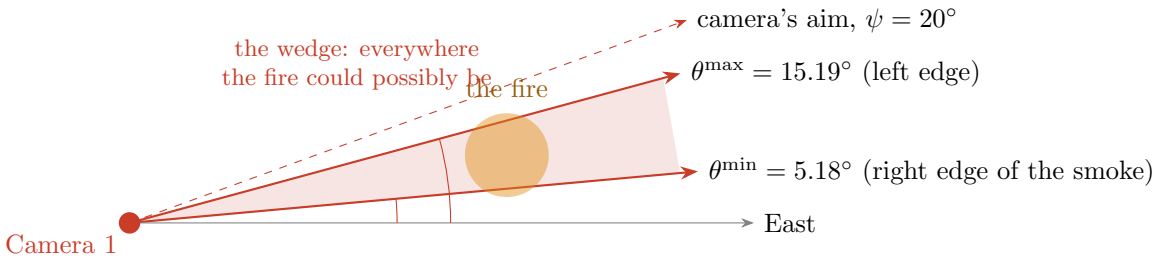


Figure 3: The smoke spans a range of pixel columns, so it spans a range of bearings. The fire is somewhere inside this wedge — but we still have no idea how far along it.

3.7 How accurate is a direction? (And what that costs in metres)

Suppose the smoke-detecting software can locate an edge to within about 1.5 pixels. How much is that in degrees, and then in metres on the ground?

Near the middle of the picture the triangle is thin, and a small sideways move of σ_u pixels tips the angle by roughly

$$\sigma_\theta \;\approx\; \frac{\sigma_u}{f} \text{ radians} \qquad \text{(for our camera: } \tfrac{1.5}{1662.77} = 0.00090 \text{ rad} = 0.052°\text{)}. \tag{7}$$

That is a tiny angle. But an angle spreads out with distance, like a flashlight beam. At a distance r the sideways uncertainty on the ground is

$$\boxed{\text{sideways error} \;\approx\; r \times \sigma_\theta \;\approx\; \frac{r\,\sigma_u}{f}} \tag{8}$$

distance to the fire	500 m	1 km	2 km	5 km	10 km
sideways error (1.5 px, $f = 1662.77$)	0.45 m	0.90 m	1.80 m	4.5 m	9.0 m

Sideways, the camera is remarkably good — better than a metre at a kilometre. Along the line of sight it knows *nothing*. This lopsided accuracy is the key fact that shapes everything in Parts 2 and 3.

3.8 A note on lens distortion

Cheap and wide lenses bend straight lines, especially near the edges of the frame, so a pixel is not exactly where the ideal triangle says. Camera engineers measure this once and publish correction numbers $(k_1, k_2, \ldots)$ that nudge each pixel back to where it belongs before you apply (2).

For a narrow, high-quality fire-watch lens the correction is often less than half a pixel and can be skipped. For a wide-angle lens it absolutely cannot: at 90° field of view, ignoring distortion can throw the bearing off by more than a degree, which is 90 metres at 5 km. We will assume our example lenses are already corrected.

Worked example — Part 1

We have $f = 1662.77$ and $c_u = 960$ for all three cameras. Take Camera 1's left smoke edge, at column $u = 1100$, with heading $\psi = 20°$.

Step 1: how far from centre?

$u - c_u = 1100 - 960 = 140$ pixels.

Step 2: the ratio.

$\dfrac{140}{1662.77} = 0.084197.$

Step 3: the angle.

$\alpha = \tan^{-1}(0.084197) = 4.8128°$. (Positive, so it is right of the camera's aim.)

Step 4: the bearing.

$\theta = \psi - \alpha = 20° - 4.8128° = \mathbf{15.1872}°$.

Repeating that for all six smoke edges:

Camera	edge	u	$(u - c_u)/f$	$\alpha = \tan^{-1}(\cdot)$	$\theta = \psi - \alpha$
1	left	1100	+0.084197	+4.8128°	**15.1872°**
1	right	1400	+0.264619	+14.8218°	**5.1782°**
2	left	700	−0.156366	−8.8871°	**78.8871°**
2	right	900	−0.036084	−2.0666°	**72.0666°**
3	left	880	−0.048113	−2.7545°	**222.7545°**
3	right	950	−0.006014	−0.3446°	**220.3446°**

So the three wedges are

$$\text{Cam 1: } 5.18° \to 15.19°, \quad \text{Cam 2: } 72.07° \to 78.89°, \quad \text{Cam 3: } 220.34° \to 222.75°.$$

The middle of each wedge.

This is our best single direction to the fire, and it is just the average of the wedge's two edges:

$$\theta_1 = \frac{5.1782 + 15.1872}{2} = 10.1827°, \qquad \theta_2 = 75.4769°, \qquad \theta_3 = 221.5495°.$$

Hold on to these three numbers — Parts 2 and 3 are built on them.

Try it yourself #1. A camera has $f = 1662.77$, $c_u = 960$, and is aimed due East ($\psi = 0°$). A smoke plume appears at pixel column $u = 1200$.

1. What is $u - c_u$?
2. What is α?
3. What is the bearing θ? Is the fire north or south of due East?
4. If the fire turns out to be 3 km away, roughly how many metres of sideways uncertainty does a 1.5-pixel detection error cause?

(Answers in Appendix C.)

4 Part 2 — Two cameras: where the lines cross

Part 1 gave each camera a straight line on the map. Two straight lines that are not parallel cross at exactly one point. Finding that point is a problem from algebra class: *two equations, two unknowns.*

4.1 Writing down the two equations

From (5), each camera gives a line through its own position with slope $m = \tan\theta$:

$$\text{Camera 1:} \qquad y = y_1 + m_1\,(x - x_1), \qquad m_1 = \tan\theta_1, \tag{9}$$

$$\text{Camera 2:} \qquad y = y_2 + m_2\,(x - x_2), \qquad m_2 = \tan\theta_2. \tag{10}$$

The fire is the one point (x, y) that is on *both* lines.

4.2 Solving them

Both right-hand sides equal y, so they equal each other:

$$y_1 + m_1(x - x_1) \;=\; y_2 + m_2(x - x_2).$$

Multiply out:

$$y_1 + m_1x - m_1x_1 \;=\; y_2 + m_2x - m_2x_2.$$

Gather the x terms on the left and everything else on the right:

$$m_1x - m_2x \;=\; y_2 - y_1 + m_1x_1 - m_2x_2,$$

$$(m_1 - m_2)\,x \;=\; y_2 - y_1 + m_1x_1 - m_2x_2.$$

Divide, and we have the answer:

$$\boxed{x = \frac{y_2 - y_1 + m_1x_1 - m_2x_2}{m_1 - m_2}} \tag{11}$$

$$\boxed{y = y_1 + m_1\,(x - x_1)} \tag{12}$$

That is the entire calculation. Plug in x from (11) to (12) to get y.

Always check your answer. Once you have (x, y), put it into *Camera 2's* equation (10) as well. You should get the same y. It costs one line of arithmetic and catches almost every mistake.

4.3 How far away is it?

The distance from each camera to the fire — useful later, and a good sanity check — is just the distance formula:

$$r_1 = \sqrt{(x - x_1)^2 + (y - y_1)^2}, \qquad r_2 = \sqrt{(x - x_2)^2 + (y - y_2)^2}. \tag{13}$$

4.4 Two traps to watch for

Trap 1: the fire must be in *front* of both cameras. Equations (9)–(10) describe infinitely long lines running *both* ways through each camera. But a camera only sees forward. So the crossing point might land behind one of them — meaning the two cameras are not actually looking at the same fire.

How to check: walk from the camera toward the answer and confirm you are heading the right way. The direction from camera 1 to the answer is $(x - x_1,\ y - y_1)$, and the direction it is looking is $(\cos\theta_1,\ \sin\theta_1)$. They point the same way when

$$(x - x_1)\cos\theta_1 + (y - y_1)\sin\theta_1 \;>\; 0.$$

Do this for both cameras. If either fails, throw the pair out.

Trap 2: vertical lines have no slope. If a camera looks due North, $\theta = 90°$ and $\tan 90°$ is undefined — your calculator will error or return something enormous. The slope method

breaks down. Near-vertical is nearly as bad: $\tan 89.9^\circ = 573$, and such huge numbers lose precision.

The version a computer should use. To avoid Trap 2 entirely, professionals skip slopes and use this equivalent formula, which never divides by anything that blows up. With $\Delta_x = x_2 - x_1$, $\Delta_y = y_2 - y_1$, and $\gamma = \theta_2 - \theta_1$:

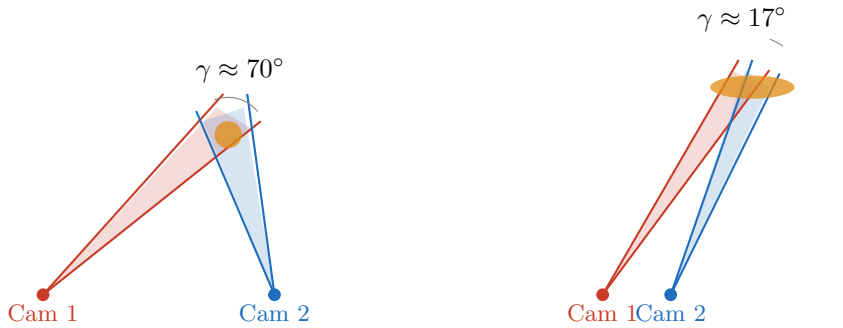
$$r_1 = \frac{\Delta_x \sin\theta_2 - \Delta_y \cos\theta_2}{\sin\gamma}, \qquad x = x_1 + r_1\cos\theta_1, \qquad y = y_1 + r_1\sin\theta_1. \tag{14}$$

It gives exactly the same answer as (11)–(12), and it hands you r_1 for free (negative r_1 means Trap 1). This is what the accompanying computer program uses. You can safely ignore it if you are doing this by hand with a well-behaved example.

4.5 The single most important thing: the crossing angle

Here is something that surprises people. The accuracy of your answer depends far less on how good your cameras are than on *where they are standing*.

Look at Figure 4. On the left, two cameras view the fire from very different directions and the lines cross cleanly, nearly at right angles. On the right, both view it from almost the same direction; the lines cross at a shallow angle and slide along each other. A tiny wobble in either angle slides the crossing point a long way.



good
a small, round uncertainty

bad
a long, smeared uncertainty

Figure 4: Same cameras, same lenses, same software — completely different accuracy. All that changed is where the cameras stand relative to the fire.

The quantity that measures this is the **crossing angle** $\gamma = \theta_2 - \theta_1$. Errors get multiplied by $1/\sin\gamma$:

crossing angle γ	5°	10°	30°	60°	90°	150°
errors multiplied by $1/\sin\gamma$	11.5×	5.8×	2.0×	1.15×	1.0×	2.0×

Rule of thumb. Keep the crossing angle between 30° and 150°. Outside that range, errors are more than doubled and the answer should be treated with suspicion.

When deciding where to put a new camera, the question is not "can it see the valley?" but "does it see the valley from a *different direction* than the cameras I already have?"

4.6 From a point to a patch of ground

Each camera actually gives a wedge, not a line. Two wedges overlap in a four-cornered shape (Figure 5) — and that shape is our estimate of the burning area.

To find its corners, notice each wedge has two edge-lines, so there are $2 \times 2 = 4$ ways to pair them up. Solve (11)–(12) four times:

Corner	is where …
P_1	Camera 1's *left* edge meets Camera 2's *left* edge
P_2	Camera 1's *right* edge meets Camera 2's *left* edge
P_3	Camera 1's *right* edge meets Camera 2's *right* edge
P_4	Camera 1's *left* edge meets Camera 2's *right* edge

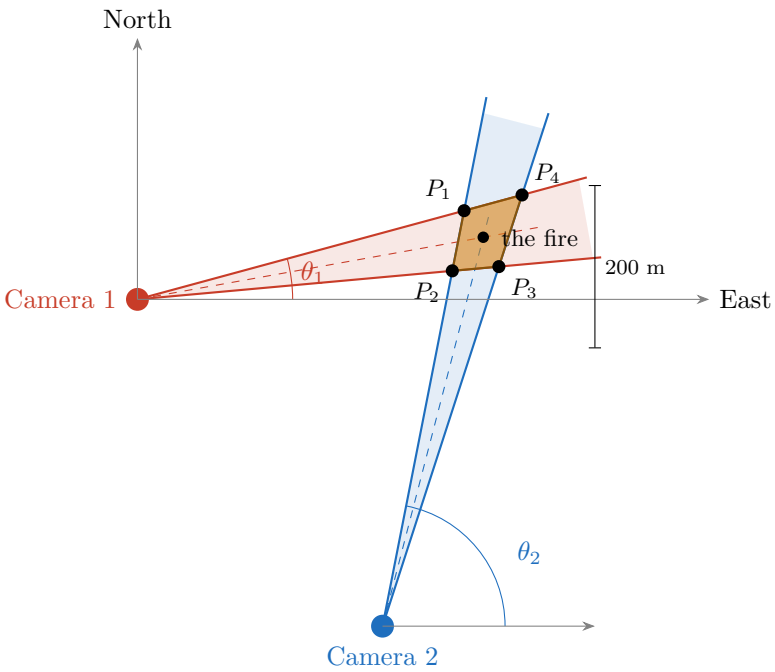


Figure 5: Two wedges overlap in a four-cornered patch (orange). Each corner is where one edge-line of Camera 1 meets one edge-line of Camera 2. The dashed middle lines cross at our single best guess for the centre of the fire. Drawn to scale from the worked example.

4.7 How much ground does that patch cover?

There is a neat formula for the area of any polygon if you know its corners, called the **shoelace formula** (because of how the multiplications criss-cross). List the corners going counter-clockwise as $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, then come back to the first one. Multiply each x by the *next* y, subtract each y times the *next* x, add it all up, and halve:

$$\boxed{\text{Area} = \tfrac{1}{2}\left|\,(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + \cdots + (x_ny_1 - x_1y_n)\,\right|} \tag{15}$$

Divide by 10 000 to convert square metres to **hectares**, the unit fire crews use.

4.8 How wrong could the answer be?

Part 1 told us each camera pins the fire down *sideways* to about $r \times \sigma_\theta$ metres. So instead of an exact line, picture each camera giving a thin *ribbon* of that half-width. Two crossing ribbons overlap in a small parallelogram: the **error box**.

The error box. If camera 1's ribbon has half-width $d_1 = r_1\sigma_1$ and camera 2's has $d_2 = r_2\sigma_2$,

the overlap is a parallelogram. Its corners sit roughly

$$\frac{d_1 + d_2}{\sin\gamma}$$

from the centre — an estimate on the generous side, so it is a safe one to quote. (For the exact corners, solve for where the ribbon edges cross, just as you did for the wedge corners.) There is that $1/\sin\gamma$ again: a shallow crossing angle stretches the box into a long splinter.

Worked example — Part 2

Cameras 1 and 2, using the middle bearings from Part 1:

$$(x_1, y_1) = (0, 0), \quad \theta_1 = 10.1827^\circ; \qquad (x_2, y_2) = (300, -400), \quad \theta_2 = 75.4769^\circ.$$

Step 1: the two slopes.

$$m_1 = \tan(10.1827^\circ) = 0.179617, \qquad m_2 = \tan(75.4769^\circ) = 3.860292.$$

Step 2: put them in (11).

$$\begin{aligned} x &= \frac{y_2 - y_1 + m_1 x_1 - m_2 x_2}{m_1 - m_2} \\ &= \frac{(-400) - 0 + (0.179617)(0) - (3.860292)(300)}{0.179617 - 3.860292} \\ &= \frac{-400 - 1158.088}{-3.680675} = \frac{-1558.088}{-3.680675} = \mathbf{423.32}\ \text{m}. \end{aligned}$$

Step 3: get y from (12).

$$y = y_1 + m_1(x - x_1) = 0 + 0.179617\,(423.32 - 0) = \mathbf{76.03}\ \text{m}.$$

Step 4: check with Camera 2.

$$y = y_2 + m_2(x - x_2) = -400 + 3.860292\,(423.32 - 300) = 76.03\ \checkmark$$

So the fire is at (423.32, 76.03) metres.

Its distances are

$$r_1 = \sqrt{423.32^2 + 76.03^2} = 430.09\ \text{m}, \qquad r_2 = \sqrt{123.32^2 + 476.03^2} = 491.75\ \text{m},$$

both comfortably in front of both cameras.

The crossing angle.

$\gamma = 75.4769^\circ - 10.1827^\circ = 65.29^\circ$, and $\sin\gamma = 0.9085$, so errors are multiplied by only $1/0.9085 = 1.10$. This is a well-placed pair.

The four corners.

Solve (11) four more times, once for each pairing of edges:

Corner	edges used	θ_1, θ_2	position (m)
P_1	left, left	15.1872°, 78.8871°	(399.89, 108.55)
P_2	right, left	5.1782°, 78.8871°	(385.43, 34.93)
P_3	right, right	5.1782°, 72.0666°	(442.43, 40.09)
P_4	left, right	15.1872°, 72.0666°	(470.82, 127.81)

The area.

Apply the shoelace formula (15), going counter-clockwise around $P_2 \to P_3 \to P_4 \to P_1$:

$$\text{Area} = 4898 \text{ m}^2 = \mathbf{0.49} \text{ hectares},$$

and the centre of that patch is (425.74, 79.12) m — only 3.7 m from the crossing of the middle lines. Those two numbers agreeing is a good sign that nothing has gone wrong.

The error box.

Each camera's ribbon half-width is

$$d_1 = r_1\sigma_1 = 430.09 \times 0.00088 = 0.38 \text{ m}, \qquad d_2 = 491.75 \times 0.00089 = 0.44 \text{ m}.$$

With $\sin\gamma = 0.9085$, the rule of thumb gives $(0.38 + 0.44)/0.9085 = 0.90$ m, and working out the four corners exactly puts the farthest one 0.76 m from the centre, with the box only 0.49 m across the other way. **We have located a fire nearly half a kilometre away to within about a metre**, using two photographs and high-school algebra.

Try it yourself #2. Camera A sits at $(0, 0)$ and sees a fire at bearing $\theta_A = 45°$. Camera B sits at $(100, 0)$ and sees the same fire at bearing $\theta_B = 135°$.

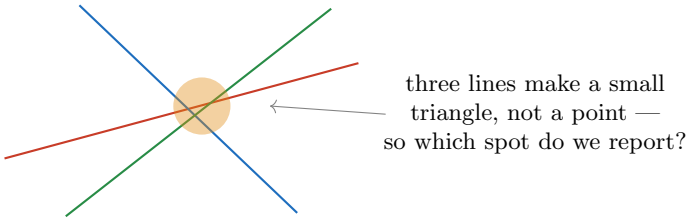
1. Find the two slopes m_A and m_B.
2. Use (11) and (12) to find the fire.
3. How far is it from each camera?
4. What is the crossing angle γ? Is this a good pair of camera positions?

(Answers in Appendix C.)

5 Part 3 — Three or more cameras

With two cameras there is always exactly one crossing point, and the math is finished. Add a third camera and something annoying happens: **the three lines do not meet at a single point.**

They miss by a little. That is not a mistake — it is guaranteed. Bearings carry small errors, camera headings are never surveyed perfectly, and the three cameras are each looking at a slightly different part of a fire that has real width. Three lines passing exactly through one point would be a miracle, not a measurement.



This is actually good news. The extra camera gives us something two cameras never had: a way to *check the answer* and to *notice when one camera is lying.*

5.1 Measuring how badly a point misses a line

To pick the best point, we first need to measure how far a candidate point (x, y) is from a camera's line. We want the **perpendicular** distance — the shortest way from the point to the line.

Camera i sits at (x_i, y_i) looking along bearing θ_i. The perpendicular distance from a point (x, y) to that line is

$$\boxed{\text{miss}_i \;=\; -\sin\theta_i\,(x - x_i) \;+\; \cos\theta_i\,(y - y_i)} \tag{16}$$

This formula does two jobs at once:

- Its **size** is the distance in metres from the point to the line.
- Its **sign** tells you *which side*: positive means left of the camera's line of sight, negative means right, and zero means the point is exactly on the line.

That sign is genuinely useful later, so keep it.

Where (16) comes from. The camera looks along the direction $(\cos\theta_i,\ \sin\theta_i)$. Turn that arrow a quarter-turn counter-clockwise and you get $(-\sin\theta_i,\ \cos\theta_i)$, which points *sideways* across the line. Measuring how far our point lies from the camera *in that sideways direction* is exactly the perpendicular distance — and that is what (16) computes.

5.2 Not all misses are equally bad

Here is the subtle point of Part 3. Suppose two cameras each miss a candidate point by exactly 1 metre. Are they equally wrong?

No. Recall from Part 1 that a camera's sideways accuracy is about $r \times \sigma_\theta$ metres, which *grows with distance*. Call this its **wiggle room**:

$$\text{wiggle}_i \ = \ r_i \times \sigma_\theta \qquad \text{(how far off this camera is normally allowed to be).} \tag{17}$$

A camera 400 m away has about 0.35 m of wiggle room, so missing by 1 m is a serious disagreement. A camera 5 km away has about 4.5 m of wiggle room, so missing by 1 m is a perfect result.

So we should not compare misses in metres. We should compare them in **how many wiggle-rooms** they are:

$$\text{score}_i \ = \ \frac{\text{miss}_i}{\text{wiggle}_i}. \tag{18}$$

Do not let this confuse you. Faraway cameras are *more forgiving* when we judge them, and also *less useful* when we average them. Both follow from the same fact: their sideways error in metres is bigger. Dividing by wiggle room handles both correctly and automatically.

5.3 Finding the best point

Now we can say exactly what "best" means:

The best point is the one that makes the total of all the squared scores as small as possible. That is, minimise

$$\text{Total} \ = \ \text{score}_1^2 + \text{score}_2^2 + \cdots + \text{score}_N^2.$$

Why squared? Squaring makes every term positive (so misses to the left cannot cancel misses to the right), and it punishes one big miss more than several small ones — which is what we want.

To write this down, give camera i three shorthand numbers computed straight from what we know:

$$a_i = -\sin\theta_i, \qquad b_i = \cos\theta_i, \qquad c_i = a_i x_i + b_i y_i, \tag{19}$$

so that (16) becomes simply $\text{miss}_i = a_i x + b_i y - c_i$. Also give it a **weight**, which is just "one over its wiggle room squared" — big for cameras we trust, small for cameras we don't:

$$w_i \ = \ \frac{1}{(\text{wiggle}_i)^2} \ = \ \frac{1}{(r_i\,\sigma_\theta)^2}. \tag{20}$$

Minimising the total (this is the one step we ask you to take on trust — it is ordinary curve-fitting, the same idea as a line of best fit) leads to two equations in the two unknowns x and

y:

$$\boxed{\begin{aligned} S_{aa}\,x \;+\; S_{ab}\,y &= S_{ac} \\ S_{ab}\,x \;+\; S_{bb}\,y &= S_{bc} \end{aligned}} \tag{21}$$

where each S is just a sum over all the cameras:

$$S_{aa} = \sum w_i a_i^2, \quad S_{ab} = \sum w_i a_i b_i, \quad S_{bb} = \sum w_i b_i^2, \quad S_{ac} = \sum w_i a_i c_i, \quad S_{bc} = \sum w_i b_i c_i. \tag{22}$$

Two equations, two unknowns — exactly like Part 2. Solving them gives

$$\boxed{x = \frac{S_{bb}S_{ac} - S_{ab}S_{bc}}{S_{aa}S_{bb} - S_{ab}^2}, \qquad y = \frac{S_{aa}S_{bc} - S_{ab}S_{ac}}{S_{aa}S_{bb} - S_{ab}^2}.} \tag{23}$$

Notice this works for *any* number of cameras. Ten cameras produce longer sums, but still just two equations at the end.

The chicken-and-egg problem, and how to escape it. The weights (20) need the distances r_i — but you cannot measure a distance until you know where the fire is, which is what you are trying to compute.

The way out is to go around twice:

1. Start with every weight set to 1 and solve (23).
2. Use that rough answer to measure the distances r_i, and get real weights from (20).
3. Solve (23) again with the real weights.

Two or three rounds is always plenty — the answer stops moving because the weights barely depend on the exact position.

5.4 Do the cameras actually agree?

With three or more cameras you can ask a question that is impossible with two: *is this answer believable?* Add up the squared scores at your best point:

$$\boxed{\text{Agreement score} \;=\; \sum_{i=1}^{N} \left(\frac{\text{miss}_i}{\text{wiggle}_i} \right)^2} \tag{24}$$

Each camera should typically contribute about 1. Two of the cameras are "used up" fixing x and y, so with N cameras the score should come out around $N - 2$. Statisticians call this quantity chi-squared; here are the numbers you compare against:

number of cameras N	3	4	5	6	8
typical score	1	2	3	4	6
too high — investigate	> 3.8	> 6.0	> 7.8	> 9.5	> 12.6

What a high score is telling you. Something in your setup is wrong — most often a camera heading ψ that was surveyed badly, or a sign convention flipped somewhere, or one camera looking at a *different* fire. Chase the camera with the biggest individual score first.

A very low score is also a warning. It means you claimed your cameras were worse than they really are, and the error box you report will be too big.

5.5 Throwing out a bad camera

Before trusting the answer, screen the cameras:

1. **Is the fire in front of it?** Use the front check from Trap 1. If not, discard that camera.
2. **Is the answer inside its wedge?** If a camera's wedge does not contain the final point, it is looking at something else.
3. **Is any single score huge?** If one camera contributes almost all of the disagreement, and the others agree nicely among themselves, that camera is the problem.
4. **Try every pair.** With N cameras there are only a handful of pairs; solve each one with Part 2's method. The good cameras will cluster; a bad one will sit far away from the cluster and give itself away.

5.6 The shape of the fire, from many cameras

For the burning *area* rather than its centre, we go back to wedges. The fire must be inside camera 1's wedge *and* camera 2's *and* camera 3's, and so on. So the answer is whatever survives all of them.

There is a pleasantly simple way to compute this, using the sign from (16):

Trimming, one edge at a time.

1. Start with the four-cornered patch from Part 2.
2. Take one edge-line of one camera's wedge. Using (16), work out the sign for each corner of the patch — which side of that line it is on.
3. Corners on the good side stay. Corners on the wrong side are deleted. Wherever the patch's boundary crosses the line, add a new corner at the crossing point.
4. Repeat for every edge-line of every camera.

What is left is the fire region. Each camera has two edge-lines, so N cameras means $2N$ trims.

Because every trim only ever removes area, **each extra camera can only shrink the estimate, never grow it**. That is exactly what you would hope: more evidence, more certainty.

If the region disappears entirely, the cameras contradict each other — there is nowhere that satisfies all of them. Do not report a result; go find the faulty camera.

Worked example — Part 3

All three cameras now, with the middle bearings from Part 1: $\theta_1 = 10.1827°$, $\theta_2 = 75.4769°$, $\theta_3 = 221.5495°$.

Step 1: the shorthand numbers (19).

Camera	position (x_i, y_i)	$a_i = -\sin\theta_i$	$b_i = \cos\theta_i$	$c_i = a_i x_i + b_i y_i$
1	$(0, 0)$	-0.176788	$+0.984249$	0.0000
2	$(300, -400)$	-0.968047	$+0.250770$	-390.7221
3	$(900, 500)$	$+0.663267$	-0.748383	$+222.7487$

Step 2: first pass, all weights = 1.

Solving (23) gives (423.02, 76.39) m. The three lines miss this point by 0.40, 0.38 and 0.66 metres — all under a metre, which is reassuring.

Step 3: work out the wiggle room and the weights.

Distances from that first answer, and (17)–(20):

Camera	distance r_i	wiggle $= r_i\sigma_\theta$	weight $w_i = 1/\text{wiggle}^2$
1	429.89 m	0.376 m	7.060
2	491.86 m	0.440 m	5.174
3	638.01 m	0.575 m	3.023

Camera 3 is farthest away, so it has the most wiggle room and the least say in the answer.

Step 4: the five sums (22).

$$S_{aa} = 6.39903, \quad S_{ab} = -3.98506, \quad S_{bb} = 8.85779, \quad S_{ac} = 2403.565, \quad S_{bc} = -1010.907.$$

The bottom of (23) is $S_{aa}S_{bb} - S_{ab}^2 = (6.39903)(8.85779) - (-3.98506)^2 = 40.8006$, so

$$x = \frac{(8.85779)(2403.565) - (-3.98506)(-1010.907)}{40.8006} = \mathbf{423.08}\text{ m},$$

$$y = \frac{(6.39903)(-1010.907) - (-3.98506)(2403.565)}{40.8006} = \mathbf{76.21}\text{ m}.$$

Step 5: does everyone agree?

At that point the three misses are +0.218, +0.277 and +0.827 metres, so the scores are

$$\tfrac{0.218}{0.376} = 0.58, \qquad \tfrac{0.277}{0.440} = 0.63, \qquad \tfrac{0.827}{0.575} = 1.44,$$

and the agreement score (24) is

$$0.58^2 + 0.63^2 + 1.44^2 \;=\; 0.33 + 0.40 + 2.07 \;=\; \mathbf{2.80}.$$

For $N = 3$ cameras the warning threshold is 3.8, so this **passes**. But notice camera 3 alone contributes 2.07 of the 2.80 — about 74%. That is the fingerprint of a camera pulling slightly against the others. Here it is off by about 1.1 m at 638 m, which is a heading error of roughly 0.1°. Worth writing in the logbook; not worth throwing the camera out.

Step 6: cross-check with every pair.

Solve each pair on its own, using Part 2's method:

Pair	crossing angle	answer (m)
Cameras 1 & 2	65.3°	(423.32, 76.03)
Cameras 1 & 3	31.4°	(421.20, 75.65)
Cameras 2 & 3	146.1°	(423.82, 77.98)
all three together		(**423.08**, **76.21**)

All three pairs land within 2.6 m of each other, and the combined answer sits sensibly among them — closest to the best-placed pair, 1 & 2. No outlier here.

Step 7: trim the region.

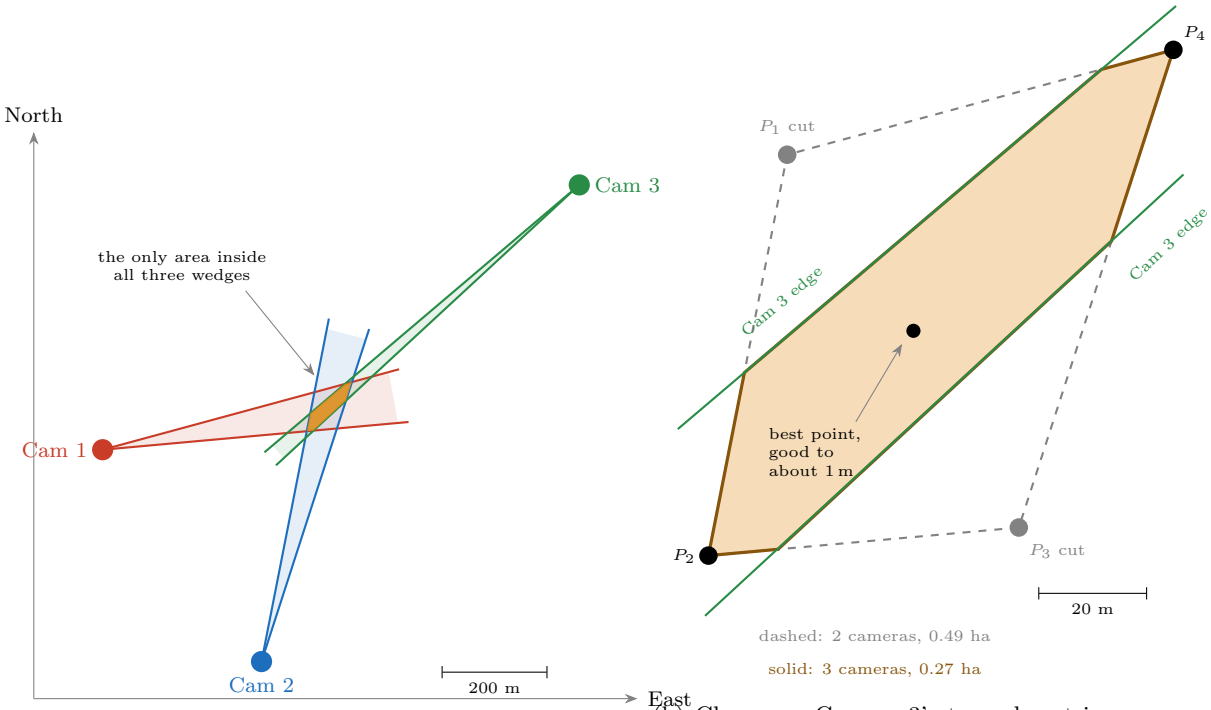
Start with the four-cornered patch from Part 2 (0.49 hectares) and trim it with camera 3's two edge-lines:

after trimming by …	shape	area
(start: cameras 1 and 2)	4 corners	4898 m^2 = 0.49 ha
camera 3's first edge	5 corners (P_1 cut off)	—
camera 3's second edge	**6 corners** (P_3 cut off)	2676 m^2 = **0.27** ha

The result.

The third camera **cut the estimated burning area almost in half**, from 0.49 to 0.27 hectares. It helped the area far more than it helped the centre point, because its wedge is narrow (2.4°) and slices right across the patch.

Finally, the best point (423.08, 76.21) does lie inside the six-cornered region — so the two calculations agree, and the result can be reported.



(a) All three wedges. The orange sliver is the only ground that all three cameras agree on.

(b) Close up. Camera 3's two edges trim corners P_1 and P_3 away, leaving six corners and about half the area.

Figure 6: The three-camera result. Note how much more precisely we know the *centre* of the fire (about a metre) than its *extent* (tens of metres) — the fire really is that big.

Try it yourself #3. A camera stands at the origin $(0, 0)$ looking along bearing $\theta = 30°$.

1. Use (16) to find how far the point (100, 60) misses the camera's line, and say which side it is on.
2. Do the same for the point (100, 50).
3. The two points are on opposite sides. Somewhere between them the line crosses $x = 100$. Find that exact y value.

(Answers in Appendix C.)

6 What actually goes wrong in real life

Every formula in this guide is exact. In a real installation, accuracy is almost never lost to the math — it is lost to *measuring the cameras*. Here is what to worry about, roughly in order of how much damage it does.

1. **The camera's heading ψ is the big one.** Get ψ wrong by 1° and every answer from that camera shifts sideways by 1.7% of the distance — that is 17 m at 1 km and **87 m at 5 km**. Worse, it is a *consistent* error, so taking more photos will never average it away.

A magnetic compass is not accurate enough (it is thrown off by declination and by the steel in the tower). Instead, photograph three or more landmarks whose map coordinates you know, and work backwards through (3) to find the ψ that fits them all best.

2. **Check your signs once, against something real.** Photograph a known landmark and confirm the computed bearing matches the map. Repeat this after *any* change to the camera or the software. A mirrored video feed or a rotated sensor silently flips (3) and produces confident nonsense.

3. **The camera's position** needs only ordinary GPS accuracy. An error here shifts the answer by about its own size, rather than being magnified by distance — so a few metres is fine.

4. **The focal length** f comes from a standard checkerboard calibration. Redo it if anyone ever refocuses or zooms the lens: on a zoom lens, f is not a fixed number.

5. **Take the photographs at the same time.** Smoke drifts. If two cameras photograph the plume ten seconds apart, you are triangulating two different objects. Keep the clocks synchronised and record a timestamp with every detection.

6. **Place the cameras thoughtfully.** Look again at Figure 4. For every part of the area you care about, make sure some *pair* of cameras sees it with a crossing angle between 30° and 150°. Check this on a map before building anything — it is much cheaper than moving a tower later.

7. **Always report the uncertainty, not just the dot.** A crew told "the fire is here, give or take 50 m" behaves very differently from one shown a single confident dot on a screen.

A The whole method on one page

You know: each camera's position (x_i, y_i), heading ψ_i, focal length f, centre column c_u; and the pixel column u_i where each one sees the fire.

PART 1 — do this for every camera separately

$$f = \frac{W/2}{\tan(\mathrm{FOV}/2)} \qquad \text{(only if } f \text{ isn't already known)}$$

$$\alpha_i = \tan^{-1}\left(\frac{u_i - c_u}{f}\right) \qquad \text{angle away from where the camera points}$$

$$\theta_i = \psi_i - \alpha_i \qquad \textbf{the bearing — the key number}$$

PART 2 — for any two cameras

$$m_1 = \tan\theta_1, \qquad m_2 = \tan\theta_2 \qquad \text{the two slopes}$$

$$x = \frac{y_2 - y_1 + m_1 x_1 - m_2 x_2}{m_1 - m_2}, \qquad y = y_1 + m_1(x - x_1) \qquad \textbf{the fire}$$

$$\gamma = \theta_2 - \theta_1 \qquad \text{keep between } 30^\circ \text{ and } 150^\circ!$$

Check the answer in camera 2's equation too. Do all four edge-pairings to get the four corners of the burning area, then use the shoelace formula.

PART 3 — for three or more cameras

$$a_i = -\sin\theta_i, \quad b_i = \cos\theta_i, \quad c_i = a_i x_i + b_i y_i$$

$$\text{miss}_i = a_i x + b_i y - c_i \qquad \text{distance from a point to camera } i\text{'s line}$$

$$w_i = \frac{1}{(r_i \sigma_\theta)^2} \qquad \text{trust nearby cameras more}$$

$$x = \frac{S_{bb} S_{ac} - S_{ab} S_{bc}}{S_{aa} S_{bb} - S_{ab}^2}, \qquad y = \frac{S_{aa} S_{bc} - S_{ab} S_{ac}}{S_{aa} S_{bb} - S_{ab}^2}$$

$$\text{score} = \sum \left(\frac{\text{miss}_i}{r_i \sigma_\theta} \right)^2 \qquad \text{should be about } N - 2$$

Go around twice: weights of 1 first, then real weights from the distances. For the area, trim the Part 2 patch with each camera's two edge-lines using the sign of miss_i.

B Symbols and words

Symbol	Say it	What it means
x, y	—	position on the map, in metres (East, North)
(x_c, y_c)	—	where a camera stands
ψ	"psi"	the camera's heading — where it is aimed
u	—	which pixel *column* something appears in
c_u	—	the middle column of the image
W	—	image width in pixels
f	—	focal length, in pixels — how zoomed-in the lens is
FOV	—	field of view: total angle the camera sees
α	"alpha"	angle from the camera's aim to the object
θ	"theta"	**bearing**: direction on the map, counter-clockwise from East
γ	"gamma"	crossing angle between two cameras' lines
m	—	slope of a line, $= \tan\theta$
r	—	distance from a camera to the fire
σ_θ	"sigma"	how much the bearing typically wobbles
σ_u	—	how many pixels the detector can be off by
wiggle room	—	$r\,\sigma_\theta$: sideways error a camera is allowed
miss	—	perpendicular distance from a point to a camera's line
w_i	—	weight: how much a camera's opinion counts
hectare	—	10 000 m^2, about two football fields

C Answers to "Try it yourself"

#1 — pixel to bearing.

1. $u - c_u = 1200 - 960 = 240$ pixels.
2. $\dfrac{240}{1662.77} = 0.144338$, so $\alpha = \tan^{-1}(0.144338) = 8.2132°$.
3. $\theta = \psi - \alpha = 0° - 8.2132° = -8.2132°$ (the same direction as 351.79°). A negative bearing is *clockwise* from East, so the fire is **south** of due East — which matches the plume appearing to the *right* of centre in the photo.
4. Sideways error $\approx r\sigma_u / f = 3000 \times 1.5/1662.77 = \mathbf{2.7\ m}$.

#2 — two cameras.

1. $m_A = \tan 45° = 1$ and $m_B = \tan 135° = -1$.

2. $x = \dfrac{0 - 0 + (1)(0) - (-1)(100)}{1 - (-1)} = \dfrac{100}{2} = 50$, then $y = 0 + 1\,(50 - 0) = 50$. The fire is at (**50**, **50**).
3. $r_A = \sqrt{50^2 + 50^2} = 70.71$ m, and by symmetry $r_B = 70.71$ m too.
4. $\gamma = 135° - 45° = 90°$, so $\sin\gamma = 1$ and errors are multiplied by exactly 1. This is the **best possible** geometry — you cannot do better than a right angle.

#3 — miss distances.

1. miss $= -\sin 30°\,(100) + \cos 30°\,(60) = -50 + 51.96 = +\mathbf{1.96}$ m. Positive, so it is to the **left** of the line of sight.
2. miss $= -50 + \cos 30°\,(50) = -50 + 43.30 = \mathbf{-6.70}$ m. Negative: to the **right**.
3. On the line, $y = x\tan\theta = 100\tan 30° = \mathbf{57.74}$. Check: $-50 + 0.866 \times 57.74 = 0$ ✓

D If you want to go further

Everything here was written with high-school algebra and trigonometry. The same ideas appear in engineering textbooks under more imposing names. If you continue in this direction, here is the translation:

What we called it	**What it is called professionally**
bearing from a pixel	the *pinhole camera model*; $(u - c_u)/f$ are *normalised image coordinates*
correcting the lens	*Brown–Conrady distortion* correction
the line from a camera	a *ray*, or a *line of position*
$(-\sin\theta,\ \cos\theta)$	the *unit normal* to the ray; the miss distance is a *dot product* with it
crossing two lines	*triangulation*; the closed form (14) is *Cramer's rule*
the $1/\sin\gamma$ penalty	*geometric dilution of precision* (GDOP); formally the *condition number*
best point from many cameras	*weighted least squares*; (21) are the *normal equations*
going around twice for weights	*iteratively reweighted least squares* (IRLS)
the error box	the *covariance matrix* and its *confidence ellipse*, found from *eigenvalues*
agreement score	the *chi-squared statistic*, with $N - 2$ *degrees of freedom*
ignoring one bad camera	*robust estimation*: the *Huber loss*, or *RANSAC*
trimming the region	*half-plane intersection*, by the *Sutherland–Hodgman* clipping algorithm

The natural next steps are to add the third dimension (so cameras can measure elevation as well as bearing, and the fire has a height), and to track the fire over time with a *Kalman filter*, which blends each new measurement with everything seen so far.

E The computer programs

Two files come with this guide:

`fire_localization_2d.py`	Every formula in this guide, written in plain Python with **no extra libraries to install**. Run it and it prints all the numbers from all three worked examples, so you can check your own arithmetic against it.
`fire_localization_2d.ipynb`	A notebook that walks through the same three parts with graphs you can change and re-run. (Needs `matplotlib`.)

```
python3 fire_localization_2d.py
```

The notebook also answers questions this guide only asserts: how badly the "shortcut" from §2.2 really behaves, how the error box stretches as the crossing angle shrinks, what a fourth camera adds, and — the most instructive one — what happens when you deliberately mis-measure a camera's heading. There you can watch the agreement score catch the mistake while the error box carries on looking reassuringly small.

All figures are drawn to scale from the worked example.
Every number in this guide is reproduced by `fire_localization_2d.py`.